

Date: 07 July 2026	Team ID: SB-7429
Project Name: Smart Lender: Applicant Credibility Prediction for Loan Approval	Maximum Marks: 5 Marks

CODE-LAYOUT, READABILITY AND REUSABILITY TEMPLATE

Code Quality Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Indentation & Style	Uniform 4-space indentations following PEP 8 standard.	Yes	Checked via linter.
2	File Layout	Modular structure separating frontend, static, views, and data scripts.	Yes	Very clean.
3	Descriptive Variables	Self-documenting labels (e.g. coapplicant_income).	Yes	Easy to read.
4	Comments & Docstrings	Explains imputation formulas, scale transformations, and Flask routes.	Yes	Thoroughly documented.
5	Modular Functions	Reusable preprocessing dictionary mappings.	Yes	DRY compliant.
6	Error Catching	Try-except wrappers inside prediction routes catch parsing errors.	Yes	Robust execution.

Reusable Components

S.No	Component Name	Language	Where Reused	Reusability Level
1	StandardScaler (scaler.pkl)	Python / Binary	Fitted during training, loaded in Flask app.py for live data scaling.	High
2	XGBoost Model (loan_model.pkl)	Python / Binary	Serialized weights called by web server for instant predictions.	High
3	Style.css	CSS	Shared styling sheet linking homepage form and results outcomes pages.	High